# MERIDIAN & ASSOCIATES LLP

Advisory | Consulting | Technology | Risk

# Supply Chain Optimization Practice Guide

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# 1 Practice Overview

Meridian's Supply Chain Optimization practice helps clients transform their end-to-end supply chains from cost centers into strategic competitive advantages. Our approach combines deep industry expertise, advanced analytics, and technology enablement to deliver measurable improvements in cost, service levels, resilience, and sustainability. Across 120+ supply chain engagements, Meridian clients have achieved average improvements of 15-25% in forecast accuracy, 12-18% in logistics cost reduction, and 20-30% in inventory optimization.

# 2 Demand Sensing and Forecasting

Traditional demand forecasting relies on historical sales data and statistical models that struggle to capture the volatility and complexity of modern markets. Meridian's demand sensing approach augments statistical baselines with machine learning models that incorporate demand signals from POS data, weather patterns, social media sentiment, economic indicators, promotional calendars, and competitive activity.

## 2.1 ML Forecasting Models

Meridian deploys ensemble forecasting models that combine multiple algorithms -- gradient boosted trees (XGBoost, LightGBM), recurrent neural networks (LSTM), and traditional ARIMA/ETS models -- to produce consensus forecasts with prediction intervals. Model selection is automated based on forecast accuracy metrics (MAPE, WMAPE, bias) evaluated on rolling holdout windows. Our models typically deliver 15-25% improvement in forecast accuracy compared to the client's existing methods, measured at the SKU-location-week level.

## 2.2 Demand Sensing Integration

Demand sensing signals are ingested through a cloud-based data pipeline (Azure Data Factory or AWS Glue) and processed in near-real-time. The sensing layer adjusts statistical forecasts based on short-term demand indicators, reducing forecast error in the 1-4 week horizon by an additional 10-15%. Integration with the client's planning system (SAP IBP, Oracle Demand Management, Kinaxis RapidResponse, or Blue Yonder) enables automated forecast updates and exception-based planner review.

# 3 Sales and Operations Planning (S&OP)

S&OP is the cross-functional planning process that aligns demand, supply, inventory, and financial plans on a rolling 18-24 month horizon. Meridian designs and implements mature S&OP processes that elevate planning from a siloed, spreadsheet-driven exercise to an integrated, executive-level decision-making forum.

- Monthly S&OP cadence: Product Review (demand plan), Supply Review (capacity and constraints), Pre-S&OP (scenario analysis, gap closure options), Executive S&OP (decisions and commitments).
- Scenario planning: What-if analysis for demand shocks, supply disruptions, capacity expansion, pricing changes, and new product launches. Quantified impact on revenue, margin, and inventory.
- Performance metrics: Forecast accuracy (MAPE by product family), plan adherence (production and shipment vs. plan), inventory health (weeks of supply, slow-moving and obsolete), customer service level (OTIF -- On-Time In-Full delivery rate, target: >95%).

# 4 Procurement and Strategic Sourcing

Meridian's procurement optimization practice covers strategic sourcing, spend analytics, supplier relationship management, and procurement technology enablement.

## 4.1 Spend Analytics

Using tools such as Coupa, Jaggaer, or custom Power BI dashboards, Meridian classifies and analyzes 100% of addressable spend to identify consolidation opportunities, maverick spending, contract leakage, and supplier rationalization candidates. Typical findings include 8-15% savings opportunities in indirect spend categories and 3-7% in direct materials through strategic sourcing events (RFPs, reverse auctions, long-term agreements).

## 4.2 Supplier Risk Scoring

Meridian's supplier risk framework evaluates suppliers across financial stability (D&B ratings, credit scores), operational reliability (on-time delivery, quality defect rates), geographic risk (single-source, country risk indices), cybersecurity posture (third-party security assessments), and ESG compliance (environmental certifications, labor practices). Each supplier receives a composite risk score that drives segmentation into strategic, preferred, approved, and conditional tiers.

# 5 Warehouse and Distribution

## 5.1 WMS Implementation

Meridian implements warehouse management systems (Manhattan Active WM, Blue Yonder WMS, SAP EWM, Oracle WMS Cloud) to optimize receiving, putaway, picking, packing, and shipping operations. Our WMS implementations include RF/barcode scanning integration, labor management modules, wave and task management, and yard management. Typical outcomes include 15-25% improvement in picking productivity, 30-40% reduction in shipping errors, and 20-30% improvement in space utilization.

## 5.2 Slotting Optimization

Slotting optimization assigns the right products to the right locations within the warehouse to minimize travel time, reduce congestion, and improve ergonomics. Meridian uses slotting optimization software (Optricity, Manhattan, or custom algorithms) to analyze order profiles, product velocity, cube utilization, and pick path efficiency. Reslotting is recommended quarterly and aligned with seasonal demand shifts.

# 6 Transportation Management

Transportation typically represents 50-65% of total logistics cost, making it the highest-impact optimization target in most supply chains.

## 6.1 TMS Implementation

Meridian implements transportation management systems (Oracle TMS, SAP TM, Blue Yonder TMS, MercuryGate) to automate carrier selection, load optimization, route planning, freight audit, and shipment visibility. TMS implementations typically achieve 12-18% reduction in transportation costs through optimized carrier selection, load consolidation, and mode conversion (e.g., LTL to FTL, parcel to LTL).

### 6.2 Route Optimization

For clients with private fleets or dedicated contract carriage, Meridian deploys route optimization solutions that consider delivery time windows, vehicle capacity constraints, driver hours-of-service regulations, fuel costs, and traffic patterns. Dynamic routing capabilities enable real-time re-optimization in response to order changes, traffic disruptions, or vehicle breakdowns. Typical outcomes include 10-15% reduction in miles driven and 8-12% improvement in stops per route.

## 7 Supply Chain Control Tower

The control tower is a centralized visibility and decision-making hub that provides real-time monitoring of the end-to-end supply chain. Meridian designs control towers with three capability layers: visibility (real-time dashboards showing order status, inventory positions, shipment tracking, supplier performance), analytics (exception detection, root cause analysis, predictive alerts for potential disruptions), and orchestration (automated response playbooks, cross-functional collaboration tools, scenario simulation). Control tower platforms include Kinaxis, o9 Solutions, E2open, and custom builds on Databricks or Snowflake.

## 8 Supply Chain Digital Twin

A digital twin is a virtual replica of the physical supply chain that enables simulation, optimization, and what-if analysis without disrupting actual operations. Meridian builds digital twins using platforms such as anyLogistix, Coupa Supply Chain Design (formerly LLamasoft), or custom Python/SimPy models. Use cases include network design (optimal number and location of warehouses), inventory policy optimization (safety stock levels, reorder points), capacity planning (make-vs-buy decisions, capital investment timing), and risk simulation (impact of port closures, supplier failures, demand shocks).

## 9 ESG and Sustainability Integration

Environmental, social, and governance (ESG) considerations are increasingly integral to supply chain strategy. Meridian helps clients measure, reduce, and report supply chain emissions and sustainability impacts.

- Scope 3 emissions tracking: Mapping and quantifying upstream (purchased goods, transportation, business travel) and downstream (product use, end-of-life) emissions using the GHG Protocol Corporate Value Chain Standard. Data collection through supplier surveys, spend-based estimation, and activity-based calculation.
- Sustainable sourcing: Supplier sustainability scorecards, preferred supplier programs for low-carbon materials, circular economy integration (recycled content, take-back programs).
- Green logistics: Mode shift analysis (road to rail/ocean), fleet electrification roadmaps, packaging optimization (right-sizing, material reduction, recyclable materials), carbon offset programs for residual emissions.
- Reporting and disclosure: CSRD, ISSB, CDP, and GRI-aligned sustainability reporting. Integration of ESG data into enterprise reporting platforms.

## 10 Resilience and Risk Mitigation

The disruptions of recent years -- pandemic, geopolitical conflicts, extreme weather events, semiconductor shortages -- have elevated supply chain resilience from a nice-to-have to a board-level priority. Meridian's resilience framework addresses four dimensions.

- Nearshoring and reshoring: Evaluating total cost of ownership (including risk premiums) for shifting production closer to end markets. Trade-off analysis between cost, lead time, quality, and resilience. Site selection support for Mexico, Eastern Europe, and Southeast Asia.
- Dual-sourcing and multi-sourcing: Identifying critical single-source components and developing alternative supplier qualification programs. Target: no more than 60% of volume for any critical component from a single supplier.
- Inventory buffers: Strategic safety stock positioning using multi-echelon inventory optimization (MEIO) models that balance service levels against inventory investment across the network.
- Contingency planning: Documented playbooks for supply disruption scenarios, demand surge events, and logistics capacity shortages. Tabletop exercises conducted annually with cross-functional participation.

## 11 Engagement Approach

Supply chain optimization engagements are structured to deliver quick wins in parallel with long-term strategic improvements. The following outlines a typical engagement:

- Timeline: 8-12 weeks for diagnostic and opportunity assessment; 6-18 months for implementation of prioritized initiatives; ongoing performance management and continuous improvement.
- Team Composition: 1 Engagement Partner, 1 Supply Chain Program Director, 2-3 Functional Leads (Planning, Procurement, Logistics), 1-2 Data Analysts (demand modeling, network optimization), 1 OCM Consultant for process change adoption.
- Key Deliverables: Current-State Assessment and Maturity Scorecard, Opportunity Register with quantified benefits, Future-State Supply Chain Design, Implementation Roadmap, Digital Twin / Network Optimization Model, Control Tower Design and Configuration, Benefits Tracking Dashboard.

## 12 Related Case Studies & Key Personnel

The following case study and key personnel are directly relevant to supply chain optimization engagements:

- Manufacturing ERP Transformation Case Study -- $4.8B global manufacturer; supply chain planning module implementation integrated with SAP S/4HANA; led by James O'Sullivan, Lead Partner. Achieved 18% reduction in inventory carrying costs and 22% improvement in OTIF delivery rates.
- Raj Krishnamurthy, Managing Director -- Data & Analytics Practice Lead with deep expertise in demand sensing, supply chain digital twins, and ML-driven forecasting. Has delivered 30+ supply chain analytics engagements across manufacturing and retail.

## 13 Related Meridian Methodologies

Supply chain engagements leverage expertise and frameworks across multiple Meridian practices:

- Standard Project Lifecycle Framework (v4.2) -- All supply chain implementations follow the SPL governance and phase gate structure.
- Organizational Change Management Framework (v5.0) -- Process redesign and technology adoption require structured OCM support, particularly for warehouse and transportation technology implementations.
- Enterprise ERP Implementation Strategy (v2.3) -- Supply chain optimization frequently coincides with or follows ERP transformation, particularly for planning (MRP/MPS) and procurement modules.

- Data, Analytics & AI Strategy Guide (v1.0) -- Advanced analytics, demand sensing, and AI-powered planning tools are central to modern supply chain strategies.